

Rudra Sharma

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PROFESSIONAL SUMMARY

Computer Science student (CGPA 8.56) specializing in Machine Learning, NLP, and Generative AI systems. Experienced in architecting LLM-powered RAG applications, optimizing ML inference pipelines, and delivering low-latency APIs. Hands-on with semantic search, vector databases, text-to-SQL pipelines, and prompt optimization. Strong problem-solving background with 400+ LeetCode solutions.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, CNNs, Model Training & Deployment, CUDA

NLP & Generative AI: LLMs, RAG Architecture, Prompt Engineering, Semantic Search, Text-to-SQL, Embeddings, Vector Databases (ChromaDB)

AI Systems: Model Context Protocol (MCP), Context Orchestration, Tool-Based Agent Communication

Backend: FastAPI, REST APIs, Pydantic

Frontend: Streamlit, HTML, CSS

Data & Analytics: Pandas, NumPy, Feature Engineering, EDA, Data Visualization

Databases: MySQL, Firebase, Supabase

Tools: Git, GitHub, Jupyter, VS Code, IntelliJ, PyCharm

Core CS: Data Structures, Algorithms, OOP, DBMS, OS Basics

EDUCATION

Bachelor of Technology in Computer Science

Krishna Institute of Engineering and Technology (KIET)
Ghaziabad, India

Expected 2028

CGPA: 8.56/10
2024 – 2028

PROFESSIONAL EXPERIENCE

E-Commerce LLM Chatbot

GitHub

- Architected an **LLM-powered RAG system** supporting semantic product search and FAQ retrieval over 1K+ records
- Implemented embedding-based retrieval with ChromaDB, improving response relevance by **35%** over keyword search
- Engineered a **text-to-SQL NLP pipeline** translating natural language queries into executable SQLite queries
- Delivered a Streamlit-based UI with average response latency under **1.2s** for real-time interactions

Credit Risk Assessment Platform

GitHub

- Designed an end-to-end ML pipeline incorporating feature engineering, SMOTE, and hyperparameter tuning
- Evaluated **5+ classification models**, selecting the best-performing model based on ROC-AUC and F1-score
- Optimized FastAPI inference endpoints to achieve **sub-200ms latency** under concurrent requests
- Integrated a responsive frontend enabling real-time risk scoring and input validation

Healthcare Premium Prediction

GitHub

- Trained regression models on structured healthcare data with **15+ engineered features**
- Performed EDA and model evaluation using MAE and RMSE metrics to guide optimization
- Delivered an interactive Streamlit application for real-time premium estimation

ACHIEVEMENTS & CERTIFICATIONS

- Achieved **99.57% validation accuracy** with optimized ML models and **sub-100ms** inference latency
- Developed and deployed multiple **LLM-based applications** using RAG, semantic search, and prompt engineering
- Solved **400+ LeetCode problems** across DSA, including trees, DP, graphs, and system design
- Certifications: Python Programming, SQL Database Management (Codebasics)